

Module: Capstone Project

Related work and data description

Assignment: SS2 – Part A

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Part A — Table 1

Publication 1

Field	Details
Title	Data-Driven Product Returns Prediction: A Cloud-Based Ensemble Selection Approach
Harvard reference	Heilig, L., Hofer, J., Lessmann, S. and Voß, S. (2016) 'Data-Driven Product Returns Prediction: A Cloud-Based Ensemble Selection Approach', <i>Proceedings of the 24th European Conference on Information Systems (ECIS)</i> , Istanbul, Turkey.
Link	"DATA-DRIVEN PRODUCT RETURNS PREDICTION: A CLOUD-BASED ENSEMBLE SELECTI" by Leonard Heilig, Julien Hofer et al.
Modelling	The study used an ensemble selection approach. The models included CART, linear SVM, logistic regression, multilayer perceptron, random forest and AdaBoost. The researchers used directed hill-climbing and bagging to create the different ensembles. The models were then compared using accuracy and AUC.
Preprocessing techniques	The authors mention that a lot of preprocessing was done before the modelling. The dataset included numerical, date and categorical variables, such as item size, colour, customer state and manufacturer. The researchers also tested different training sample sizes to see how the size of the dataset affected the prediction performance.
Datasets used	The study used the DMC 2014 competition dataset. The dataset contains e-commerce apparel orders where products were either returned or kept. The training dataset contains 481,092 observations and 14 variables.
Key takeaways	1. Ensemble methods such as Random Forest may improve return prediction performance and should be considered during model comparison for the STADIOalot project. 2. Using more training data can improve the performance of the prediction model. 3. A return prediction model can be used before or during the purchase process to help retailers reduce avoidable returns.

Part A — Table 2

Publication 2

Field	Details
Title	Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns
Harvard reference	Mishra, A. and Dutta, P. (2024) 'Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns', <i>Journal of Cleaner Production</i> , 477, 143802.
Link	Return management in e-commerce firms: A machine learning approach to predict product returns and examine variables influencing returns - ScienceDirect
Modelling	The study compared three machine learning algorithms: logistic regression, random forest and gradient boosting. The models were compared using ROC curves and AUC to determine which model performed better at predicting product returns.
Preprocessing techniques	The study prepared e-commerce transaction data for machine learning by selecting variables related to orders, products and payment methods. The researchers looked at the variables that could influence product returns and prepared the data so that the different classification algorithms could be compared.
Datasets used	An e-commerce industry dataset containing information about product orders and customer return behaviour.
Key takeaways	1. Different machine learning models can be compared to find a better model for predicting returns. 2. Product category, order information and transaction variables should be included during feature engineering because they are associated with return behaviour. 3. ROC-AUC can be useful for comparing classification models when predicting whether an order will be returned.

Part A — Table 3

Publication 3

Field	Details
Title	Understanding and predicting online product return behavior: An interpretable machine learning approach
Harvard reference	Duong, Q.H., Zhou, L., Van Nguyen, T. and Meng, M. (2025) 'Understanding and predicting online product return behavior: An interpretable machine learning approach', <i>International Journal of Production Economics</i> , 280, 109499.
Link	Understanding and predicting online product return behavior: An interpretable machine learning approach - ScienceDirect
Modelling	The study developed an interpretable machine learning approach using product information taken from customer reviews. Random forest was used to predict product return behaviour and identify product and review characteristics that were linked to a higher chance of a return.
Preprocessing techniques	The study used customer review data and extracted product attributes from the reviews. The dataset included different product categories, such as food and groceries, books, fashion, electronics and home appliances.
Datasets used	Amazon product review data covering five product categories, including food and groceries, books, fashion, electronics and home appliances.
Key takeaways	1. Product characteristics and customer review-related information may provide valuable features for identifying high-risk return items. 2. Different product categories may have different factors that influence returns. 3. An interpretable model can help explain why an item has a higher return risk instead of only providing a prediction.

Part A — Table 4: Public Dataset

Public Dataset

Part A — Table 4: Public Dataset

Field	Details
Name	Brazilian E-Commerce Public Dataset by Olist
Harvard reference	Olist (2018) <i>Brazilian E-Commerce Public Dataset by Olist</i> . Kaggle.
Link	Brazilian E-Commerce Public Dataset by Olist
Number of instances	The dataset contains approximately 100,000 orders made between 2016 and 2018. The data is divided into several related datasets containing information about orders, order items, products, customers, sellers, payments and reviews.
Number of feature columns	The dataset contains several related tables with different numbers of columns. The tables provide information about orders, products, customers, sellers, payments, reviews and delivery, which can be combined to create the features required for the analysis.
Number of classes	2 Classes (Binary Classification). The dataset contains an order_status field and a review_score field. For this project, the target variable will be engineered as a binary outcome based on customer satisfaction. Orders with review scores of 1-2 will be classified as a negative outcome, while orders with review scores of 3-5 will be classified as a positive outcome.
Target range	Not applicable because this is a classification problem.
Origin	The dataset comes from Olist, a Brazilian e-commerce company, and contains approximately 100,000 anonymised orders made between 2016 and 2018. The dataset contains order, product, customer, seller, payment, review and delivery information. The different datasets can be combined using common identifiers such as order ID, customer ID, product ID and seller ID. This information can be used to analyse e-commerce transactions and develop a machine learning model for the selected outcome.

The published description of the Olist dataset states that it contains approximately **100,000 orders** made between 2016 and 2018. The dataset includes information about orders, order items, products, customers, sellers, payments, reviews and delivery. These variables provide information about the products purchased, customer details, order status, payment methods, delivery dates and customer review scores.

The Olist dataset includes separate CSV files for orders, order items, products, customers, sellers, payments, reviews and geolocation. These datasets can be combined using common identifiers such as `order_id`, `customer_id`, `product_id` and `seller_id`. The combined data can then be used to select the relevant features for the machine learning model and develop the selected classification outcome.